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COMPUTER SCIENCE

INZULULWAZI NGEZEKHOMPYUTHA

REKENAARWETENSKAP

cs.sun.ac.za

Cornelia P. Inggs W.H.K. Bester

Programme & Project Introduction

CS797: HONOURS IN COMPUTER SCIENCE

Monday, 2 February 2026

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Honours Coordinators

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Honours/BDatSci Project Coordinator

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Questions

- Send us e-mail or a Teams message.
- Schedule a meeting.
- Speak to the Class Representative.

We prefer to sort out issues before they become problems. So, please, *don't wait* before speaking up.

To Do This Year

The Honours in Computer Science programme consists of 128 credits.

1. You MUST pass 6 semester subjects ($6 \times 16 = 96$ credits):
   - At least 4 subjects from CS;
   - More about the subject choices later.
2. You MUST pass the year project (32 credits).
   - This is the killer!
   - The project descriptions are on the web.
3. You MUST be a student assistant for at least one undergraduate CS module.
4. You SHOULD have fun!

List of modules: cs.sun.ac.za/teaching/PG
List of projects: cs.sun.ac.za/courses/cs771

To Do in February

Deadline: 6 February 2026, 17:00

- Complete the project selection form.
- Find an undergraduate course to demi—speak directly to the lecturers.

Deadline: 9 February 2026, 08:00

- Outside module requests—e-mail Cornelia.
- Register.
- Elect a class representative.

Deadline: 20 February 2026

Last day to change Semester I modules

Deadline: 31 July 2026

Last day to change Semester II modules

The Honours Lab

Seats in the Lab

You MAY choose your own dedicated seat in the honours lab.

Rules of Conduct

1. Be considerate to other people in the lab.
2. Keep it neat, and clean up after yourself.
3. You may eat and drink in the lab, but see Rule 2.
4. Don't ever jam any outside door open.
5. Arrangements for coffee will be sorted out once we move in (and get you your new coffee machine).

MENTAL HEALTH ADVICE

We strongly urge you to work in the lab whenever possible. Especially if you live outside Stellenbosch and drive in, consider the lab your office during office hours.

Honours Project
in Computer Science

Your Honours Project

- Is a compulsory 32-credit year module;
- Consists of **a substantial piece of independent software development or research;**
- Typically involves writing either nontrivial or a large amount of code (or both);
- Will be supervised by one or two lecturers;
- **But ultimately, puts you in the driving seat;**
- If done well, is an excellent addition to your resumé…
  and helps when looking for an MSc supervisor;
- Has the following deliverables at the end of the year:
  1. a project report,
  2. a project demonstration, and
  3. a project presentation;
- The module framework is available at cs.sun.ac.za/courses/cs771.

Project Focus

Software Development vs Research

Is this a well-defined project that mainly involves building a piece of software?

OR Is the goal to (in)validate some hypotheses?

OR Is it a bit of both?

Bottom Line

The final report contents differ somewhat:

§	SOFTWARE DEVELOPMENT	RESEARCH
1.	Introduction	Introduction
2.	Background & requirements	Literature survey & related work
3.	Design & implementation	Research questions/hypotheses & experimental design
4.	Testing	Results & evaluation
5.	Conclusion & future work	Conclusion & future work

These are just broad ideas… always listen to your supervisor!

Time Commitment

Notional Hours

- 32 credits = 320 notional hours $\implies$ ±12 hours per week (semester time only).
- This guideline is for passing comfortably, *not* doing well.
- Recommendation: Plan for working on the project at least 10 hours per week, and **you SHOULD work on your project during recess.**

First Semester: Weekly Project Hour

- Fridays, 11:10–12:00, Van der Sterr Building 1017.
- Hosted by Willem (and others on an ad hoc basis).
- Report writing, LaTeX, project planning, "missing CS semester" stuff, etc.
- Supervisors have different supervision styles, and this will level the playing field somewhat.

You and Your Supervisor

You

- MUST schedule regular project meetings with your supervisor;
- MUST use your assigned CS GitLab repositories for code and reports;
- MUST keep backups of all other necessary data;
- MUST write reports in LaTeX, and submit them via SUNLearn;
- **MUST work hard from Day 1, at least 10 hours per week.**

Your Supervisor

- MAY have to deviate from meeting schedules because of other responsibilities;
- MAY require you to follow a software engineering process (→ RW344);
- MAY require regular progress reports or a backlog;
- MAY deviate from these rules, but MUST NOT contradict the module framework.

What We'd Like vs Reality

WHAT WE'D LIKE...	BUT QUITE OFTEN, IN REALITY...
You will start working from Day 1.	You will wait too long to start.
You will work at least 10 hours per week, every week.	You have other modules and deadlines, and the project takes a back seat.
You will make regular progress.	You *will* get stuck at least once.*
You will work regularly throughout the holidays.	You go spelunking, binge on series, or disappear into gaming.
You will finish with time to spare, to polish your app and/or report.	You run out of time, and try to do everything the week before a deadline.

*"Plan to throw one away; you will anyhow." —Fred Brooks, *The Mythical Man-Month*, as quoted by Eric S. Raymond

For Your Sanity, Manage Your Project…

- **Schedule your work time wisely.**
  - Block daily project time, just like for any other subject.
  - Remember, context switching is expensive.
  - Keep a list of project things you can do with half your neurons on vacation, like refactoring your code, or making graphics for your report.
- **Schedule regular down time, daily and weekly.**
  - Make time for whatever keeps you sane.
  - Try not to pull all-nighters.
  - By all means, take a couple of days totally off during recess… but if this runs beyond two weeks during the winter recess, you'll probably end up in trouble.
- **Keep backups of everything. Commit your code regularly.**
- **Academic writing takes longer than you expect,** even if you've done it before. Start early, and expect to write and rewrite.
- **If there is an issue, speak up, please. IMMEDIATELY.**

2026 Project Schedule & Mark Weights

DATE	TYPE	DESCRIPTION	WEIGHT
27 Feb	F	Project brief, requirements, research questions	——
27 Mar	F	Max. 7 page draft report; see the module framework	——
15 May	S+F	DEMO 1 Max. 7 page report (as above) + demo*	10%
27–30 Jul	S+F	DEMO 2 Just demo; no report	20%
4 Sep	F	Draft final report	——
23 Oct	S	DEMO 3 Final report	30%
9–13 Nov†	S	DEMO 3 Final demo	30%
[ditto]	S	DEMO 3 Presentation	10%

F = formative (i.e., for feedback, whether written or via meeting)
S = summative (i.e., for marks)

*To be scheduled between 22 and 30 May
†May be moved one week earlier or later—subject to examiner and moderator availability

Project Assessment

Deliverables & Marking Scheme

Report Marks are based on the quality (not quantity) and clarity of the report, and the scope of the work.

Demo Show code running (app, notebooks, etc.); marks are based on the quality and scope of the work, and insight when answering questions.

Presentation Marks are based on the quality and clarity of the presentation.

There is a 2nd opportunity for Demo 3, and we won't hesitate to fail students on the 1st opportunity, in which case, graduation in December is impossible.

Postgraduate Studies: Your First Taste of "Real Life"

- Use to your benefit what you're good at, and where you can improve.
- Deliver an excellent product or excellent results.
- Convince us (by your reports, demos, and presentation).

Project Selection & Allocation

Your Choice

1. Read through the projects posted at cs.sun.ac.za/courses/cs771.
2. You SHOULD discuss the projects you find promising with their supervisors.
3. You MAY propose a project topic, but you need to find a willing supervisor, and each supervisor can supervise at most one student-proposed project.
4. Project availability MAY change until 5 Feb, 09:00, when selection opens.
5. You MUST rank between 7 and 10 top choices by 6 Feb, 17:00.
   - You MUST NOT choose more than 2 projects from Steve (SK) or more than 3 projects from any other CS academic.
   - You MUST choose from at least 4 different supervisors.

Allocation

We strive to be fair, but cannot guarantee you get a Top 3 choice. The project allocations will be published at the end of next week.

Honours Modules in Computer Science

Honours Modules

- You MUST offer 96 credits = 6 CS Honours modules.
- You MAY offer at most 2 modules from related departments.
  - Math and Applied Math are typical choices and work well, depending on your academic strengths and history.
  - But be careful: Such modules have their own prerequisites and different timetables, and some modules may be worth fewer credits.
- We *strongly urge* you to balance your choice to 48 credits (= 3 CS Honours modules) per semester; if you do overload, frontload the first semester.
- Some undergraduate modules are available as honours modules:
  - You MUST NOT register for a module you passed as undergrad;
  - You MUST take Networks (14530-711) if you did *not* take CS313 as an undergrad;
  - You MUST take Concurrency (64963-714) if you did *not* take CS314 as an undergrad;
  - You MAY take Machine Learning A (14195-742 ≡ CS315) without having done Mathematical Statistics;
  - You MAY take Automata Theory (63452-711) if you did *not* take CS345 as an undergrad.

CS Honours Module Catalogue 2026, Semester I

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CODE	DESCRIPTION	LECTURER(S)
64947-712	Advanced Algorithms	Willem Bester
14232-791	Computational Intelligence [Artificial Intelligence]	Andries Engelbrecht
14530-711	Computer Networks 1 ≅CS313	Bill Tucker
14533-771	Computing & Society	Bill Tucker
64963-714	Concurrent Programming 1 ≅CS314	Cornelia Inggs
64572-793	Digital Image Processing APPLIED MATHS	Hanno Coetzer
13944-795	Functional Programming	Brink van der Merwe
14195-742	Machine Learning A ≅CS315	Johan du Preez & Trienko Grobler & Steve Kroon
65048-746	Ontology Engineering [Advanced Topics in CS 2]	Maria Keet*
64971-716	Vulnerability Discovery & Exploitation [Advanced Topics in CS 1]	Fabian Yamaguchi‡

*R&D Manager, meaningfy.ws, PhD (Bolzano) ‡Founder & CTO, whirlylabs.com, PhD (Göttingen)

CS Honours Module Catalogue 2026, Semester II

CODE	DESCRIPTION	LECTURER(S)
63452-711	Automata Theory & Applications ≅CS345	Brink van der Merwe
14888-743	Cognitive Robotics [Machine Learning & Artificial Intelligence I]	Gavin Rens
65021-745	Compilers [Software Construction]	Bernd Fischer
62847-792	Computer Vision APPLIED MATHS	Shane Josias
11788-741	Machine Learning	Andries Engelbrecht
14531-771	Principles of Data Science	Marcel Dunaiski & Mkhuseli Ngxande
14065-796	Program Analysis	Bernd Fischer & Cornelia Inggs
14066-791	Space Science Algorithms	Trienko Grobler

Maria Keet's Module

Ontology Engineering (65048-746)

The principal aim of this module is to provide the participant with an overview of ontology engineering – including knowledge representation languages, automated reasoning, and methods and techniques for ontology development – and main application fields. Ontologies are used in a wide range of applications and back-end computing tasks, such as data integration, information retrieval, and natural language processing. The topics covered include:

- Logic foundations for ontologies: Languages and Automated reasoning;
- Ontology development: Ontology engineering top-down & bottom-up;
- Methodologies for ontology development and maintenance, methods to enhance ontology quality;
- a selection of current research topics.

www.meteck.org/teaching/OEbook

Choosing Your Honours Modules

- Make your choice to register before classes start (necessary to access SUNLearn).
- If you are uncertain, attend the first week's contact sessions for those modules in which you are interested.
- Make any changes to your initial module choices before the deadline.

MODULE	FIRST LECTURE	VENUE
Advanced Algorithms	Mon 9 Feb, 11:10	MathSci/IndPsyc 2002
Computational Intelligence	Thu 12 Feb, 13:10	GenEng A3011
Computing & Society	Mon 9 Feb, 13:10	VdSterr 2118
Digital Image Processing	Wed 11 Feb, 10:10	VdSterr 2118
Functional Programming	Mon 9 Feb, 09:10	InorgChemistry 2017
Machine Learning A ≅CS315	Mon 9 Feb, 08:10	Merensky 1011
Ontology Engineering	Fri 13 Feb, 10:10	VdSterr 1026
Vulnerability Discovery & Detection	Fri 13 Feb, 12:10	VdSterr 2118

Any last questions?

MODULES

PROJECTS